Lab 1: Setup and Learning Basic Shell Commands

In this lab, we're setting up access to a central server that we're using during the semester to do the labs on. You can connect to it with your st-account via the SSH (secure shell) protocol from your own machine. You need to connect to the h-da VPN from the outside or directly from the university network. On the server, you have your own home directory. This is where you can work on the labs and create files that contain your solutions. You'll then clone a GitLab repository into your home directory to submit your solutions. Follow these instructions to set everything up for this and later lab assignments.

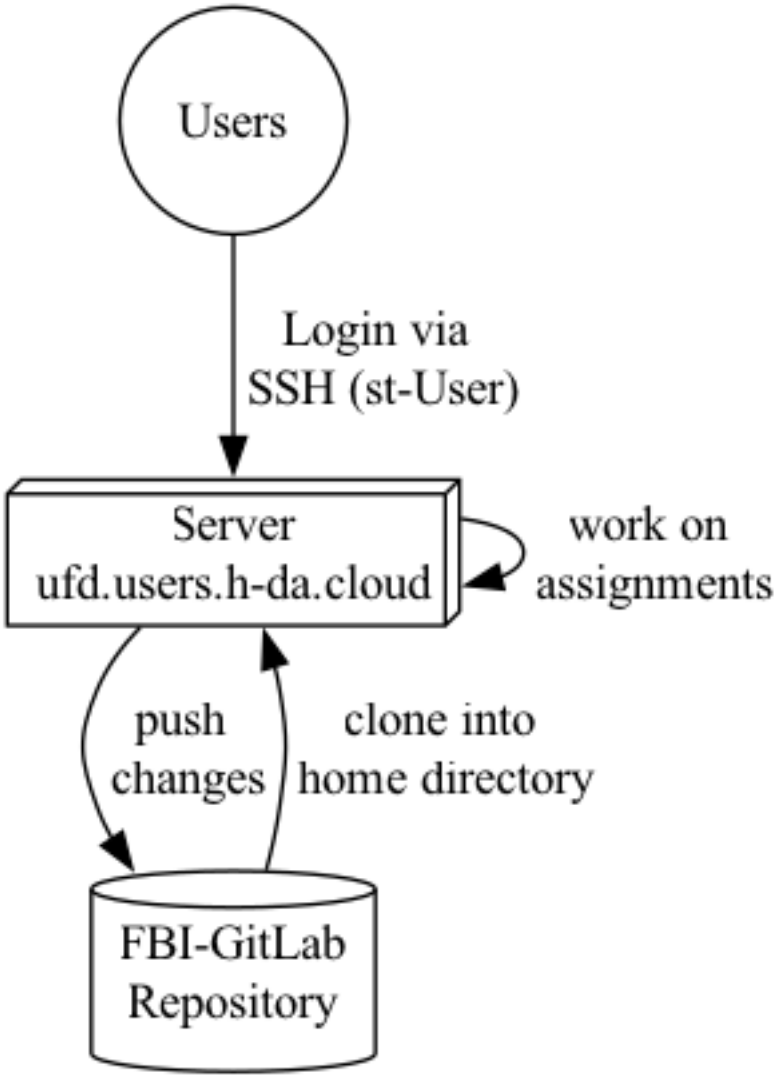


Figure 1: Lab Infrastructure

Setup Instructions

In this guide, you will learn how to:

1. Log into the central UFD server: `ufd.users.h-da.cloud`
2. Create an SSH key using the ed25519 algorithm in your home directory.
3. Upload the SSH public key to your GitLab account.
4. Load the SSH key into the keychain for seamless authentication.
5. Clone your own personal GitLab repository to your home directory on the server using SSH authentication.

Part 1: Logging into the UFD Server

Step 1: Open a Terminal on Your Local Machine

A terminal or terminal emulator is available for various operating systems. Unix systems (including Mac OS) have them built-in. Windows systems have the powershell. Ensure that there is an ssh client available. A popular client for Windows is PuTTY (`https://putty.org/`). Unix systems can use kitty (`https://sw.kovidgoyal.net/kitty/`) or other clients already installed. If there is no ssh client available, install the `openssh` package to get it (use your distribution's package manager to search for it).

Use the terminal on your own computer to SSH into the central UFD server called `ufd.users.h-da.cloud` (IPv6 only). If you haven't connected before, use the following command, replacing your-username with your own st-user credentials:

```
ssh your-username@ufd.users.h-da.cloud
```

Confirm the SSH host fingerprint, which uniquely identifies this host (only upon the first connection). You'll then asked for your password to your st-user ID. Once entered correctly, you're connected to the server and put into your home directory.

Part 2: Creating an SSH Key on the UFD Server

Step 1: Generate the SSH Key

In this part, we generate an extra key to connect to your UFD repository on GitLab. While logged into the FreeBSD server, use the following command to generate an SSH key with the `ed25519` algorithm in your home directory:

```
ssh-keygen -t ed25519 -C "UFD key for <yourusername>"
```

Replace “<yourusername” with your actual username.

When prompted to choose a file to save the key, press Enter to use the default location.

Example:

```
Enter file in which to save the key (/home/yourusername/.ssh/id_ed25519):
```

When asked for a passphrase, you can either press Enter to leave it empty or set a passphrase for added security. Remember your passphrase! If you forget it, you have to generate a new key.

Step 2: Copy Your Public Key into GitLab

After generating the key, display the contents of the public key file using the following command:

```
cat ~/.ssh/id_ed25519.pub
```

The output should look similar to this:

Example:

```
ssh-ed25519 AAAAC3NzaC1lZBI1NTE5AAGAIKdqQbhG7bCDvbQn+OyRoZmk/dyNcZQfLQ UFD
↪  key for <yourusername>
```

Copy the entire line, starting with ssh-ed25519 and ending with your username.

Part 3: Uploading the SSH Key to GitLab

Once your SSH key is generated, you need to upload the public key to GitLab. Do this step on your machine in a browser.

Step 1: Log into GitLab

1. Go to the FBI GitLab instance (`https://code.fbi.h-da.de`).
2. Log into your account if you have not already done so.

Step 2: Add Your SSH Key

1. In GitLab, click on your profile icon.
2. Select *Preferences*.
3. In the User settings sidebar, click on *SSH Keys*.
4. Click on *Add new key*
5. Paste your SSH public key (copied earlier) into the Key field. In the Title field, give your key a meaningful name, such as “UFD Server Key”. Optionally, set an expiration date for your key. Leave the other fields as they are.
6. Click the *Add Key* button.

Congratulations! Your SSH key is now linked to your GitLab account.

Part 4: Loading the SSH Key into the SSH Agent

In order to use your SSH key for GitLab operations, you'll need to load the key into the SSH keychain on the UFD server. The keychain protects the key in memory and handles communication between it and services that require the key for authentication (GitLab in our case). If you set a passphrase for the key, it will be used when required so that you don't have to type it every time. The agent takes care of that.

Step 1: Start the SSH Agent Run the following command to start the SSH agent (if it's not already running):

```
eval "$(ssh-agent -s)"
```

This will return the process ID (PID) of the ssh-agent process to confirm it is running. This is only valid for this session. As soon as you log out of the server and log back in, you have to start the agent and load the key again.

Step 2: Add Your SSH Key to the Agent Next, add your newly created SSH key to the agent using the following command:

```
ssh-add ~/.ssh/id_ed25519
```

If you set a passphrase when creating the SSH key, you will be prompted to enter it at this point.

Step 3: Verify the SSH Key is Loaded To ensure your key has been added to the agent, run:

```
ssh-add -l
```

This will list the keys currently loaded. You should see the key you just added listed with its fingerprint. Well done!

Part 5: Cloning the UFD GitLab Repository to Your Home Directory

Now that your SSH key is linked to GitLab and loaded into the SSH agent, you can clone repositories to your home directory on the server.

Step 1: Find the SSH URL for the Repository

1. Go to the GitLab project repository you want to clone.
2. Click on the Clone button and copy the SSH URL. It will look something like this:

Example:

```
git@gitlab.example.com:your-username/your-repository.git
```

Step 2: Clone the Repository

While still logged into the UFD server, navigate to your home directory (if not already there) by entering `cd` and use the `git clone` command with the SSH URL:

Example:

```
git clone git@gitlab.example.com:your-username/your-repository.git
```

This will clone the repository into your home directory. You need to do this only once. In future operations, you need only update (`git pull`) the repository.

That's it, you should now have your repository in your home directory on the UFD server. You can use it throughout the whole semester to submit your lab results.

Learning Basic Shell Commands

Unix uses a couple of very common commands to navigate the file system and to perform work. We're going to explore and learn about some of these commands as they form the basis for a lot of other Unix actions.

Lab 1 Objectives

As part of your repository, you'll find that the `Lab1` directory contains a directory called `Game`. Change into that directory using the terminal using `cd`.

Note: The game files work only on the UFD server and may not on other Unix systems. This is known and somewhat intentional.

Your objective for this lab is to play this game that teaches you some Unix commands along the way. The `README` in this directory provides instructions on how to play it. Have fun! If you want to play the game again, remove the entire directory from your home directory and run `git pull` again to retrieve the files again from the repository.

- Make a list of commands (like `ls`) that you used during the game.
- Create a dungeon map that shows which rooms connect to others. List what the room contains initially (not after manipulating any of its contents).
- Riddle specific questions:
  - What is the relationship between the altar in the chapel and the courtyard fountain?
  - How did you create a potion in the nursery?
  - From which room did you create the portal to get into the rift?

  - How did you find out the robot activation sequence?
  - How many hit points (HP) did you have at the end of the game?
  - What was in your inventory when the game ended?

- Which commands did you learn about (that were new to you) by playing the game?
- Which commands were confusing or difficult to understand for you?
- What kind of aliases would you define for your own shell usage to make navigating easier?

Lab submission:

Answer these questions by filling out the `Lab1-docs.md` file in your repository. On the UFD host you can use the easy editor `ee` followed by the filename. When you're finished, `git add` the file, followed by `git commit` and `git push` to submit your results into the repository.

Note: Do this before or during (at the latest) your regular lab date. Show your results for a final evaluation. You have to show up to the lab, pushing the file to the repo is not enough to get this lab marked as completed.